

Chapter 17 - Carbohydrates

- General Formula $(\text{CH}_2\text{O})_n$ such as $\text{C}_6\text{H}_{12}\text{O}_6$ which is the formula for glucose
mono- di- oligo- and polysaccharides are possible
- Names generally end in *-ose* aldose (if aldehyde based), ketose (if ketone based)
number of C's triose, tetrose, pentose, hexose, etc. or combos: aldopentose, ketohexose
- Stereochemistry: left- and right-handed pairs (enantiomers), non-superimposable mirror images, we say these molecules (or objects) are *chiral*; four different groups on C;
by convention, in a Fischer projection, the position of the -OH second from the end (lowest cross-piece) determines the designation: if -OH points to the right, it's D-
if -OH points to the left, it's L- e.g., *D-glucose*
- Ring structures are more stable than open-chain saccharides. The C_1 (aldehyde) closes up with the O from the hydroxyl group of the C_5 to form a **five-membered ring (= furanose)** or a **six-membered ring (= pyranose)**. Be able to recognize an α (down) versus a β (up) linkage at the anomeric C in a Haworth projection.
- Specific saccharides: KNOW
monosaccharides: **fructose**: 5-membered ring, "fruit sugar", sweetest, corn syrup, $\frac{1}{2}$ of honey
galactose: hexose similar to glucose, half of the disaccharide of lactose (milk sugar)
ribose: found in RNA; the deoxy- form is a major component of DNA
glucose: formula is $\text{C}_6\text{H}_{12}\text{O}_6$; "blood sugar" or dextrose; result of photosynthesis
disaccharides: held together by a **glycosidic bond** (α (down) or β (up), see p. 567)
maltose - glucose + glucose $\alpha(1 \rightarrow 4)$ ("malt sugar")
cellobiose - glucose + glucose $\beta(1 \rightarrow 4)$ (cellobiose comes from cellulose)
lactose - galactose + glucose ("milk sugar" - all mammals)
sucrose - glucose + fructose ("cane sugar", "table sugar", "beet sugar")
the glycosidic bond in sucrose is different from other mono- and disaccharides
Thus, sucrose is a non-reducing sugar. All monosaccharides and all other disaccharides are reducing sugars : will reduce the Cu^{+2} of Benedict's Reagent from blue solution to a red-orange solid.
- Polysaccharides: - large number of saccharide rings - linked through O atoms= **glycosidic bond** Starch (A,B,C as described below, but not D)
(A) amylose- $(\alpha\text{-D-glucose})_n$ - linear polymer of glucoses, coiled into a helix, unbranched
(B) amylopectin- $(\alpha\text{-D-glucose})_n$ - not linear, but somewhat branched $\alpha(1 \rightarrow 6)$
(C) glycogen, very much like amylopectin, but even more highly branched $\alpha(1 \rightarrow 6)$;
short term energy storage sometimes called "animal starch"; found in the liver
(D) cellulose - most abundant, $(\beta\text{-D-glucose})_n$; plant cell walls; indigestible by humans; fiber
- what is reducing, what is not a reducing sugar; what will react + with Benedict's reagent?
Benedict's is positive with: all monosaccharides, most disaccharides (EXCEPT sucrose), α -hydroxy aldehydes, and α -hydroxy ketones (middle of page 570, p.467); NOT ketones
- What would a positive reaction with Benedict's reagent look like? (p. 570)
- Smaller carbohydrates are *very* water soluble due to H-bonding with polyhydroxy groups.
- Polysaccharides may form a *suspension*, colloid-like pseudo-solutions.

Chapter 18 Lipids

1. What is a lipid? How is it defined?
2. What are saponifiable and non-saponifiable lipids?
3. The simple saponifiable lipids include triglycerides and waxes. Check out their structures. You will be responsible for looking at a structure and identifying its class. What makes a wax different from a triglyceride?
4. The triglyceride structure (and its related molecules) is what you may have to draw. Know the schematic structures (cartoons) such as on p. 593. Be able to **draw** the triester as on p. 587.
5. The reaction map on p. 592 is useful. Study it. You need to be able to assemble/disassemble a triglyceride from glycerol and 3 f.a.; show hydrolysis to form glycerol and 3 f.a. salts ("soaps"); and show the reduction with hydrogen to change the C=C to all C-C in the f.a.
6. Complex saponifiable lipids include sphingolipids and phosphoglycerides. Study their structures to be able to differentiate them. They both contain P atoms but phosphoglycerides are much more similar to triglycerides in structure. Be able to pick the correct structure from a list.
7. The non-saponifiable lipids include steroids and prostaglandins. Steroids are made up of four fused rings (6,6,6, and a 5). Know some examples (from text and notes) and their functions. Prostaglandins are hormone-like molecules with bobby-pin shapes. Know their functions and be able to pick them out from a series of structures.
8. Fatty acids (f.a.) are components of triglycerides. Their mp goes up with increasing C's; their mp goes down (dramatically) with increasing numbers of C=C. Be able to list in order of mp.
9. Know the important aspects of saturated and unsaturated fatty acids; and fat (solids) and oils (liquids). Plant and animal sources. Generalizations and health considerations.
10. How does soap work? What is a micelle? Be able to draw it and explain the action.

Chapter 19 Proteins

1. A protein is a linear polymer of amino acids (AA). A polypeptide is the same thing only with a smaller number of AA units.
2. Know the general structure of amino acids and the conventional ways they are drawn. You will be given a handout copy of p. 613 or equivalent. Know what a zwitterion is.
3. What is the isoelectric point (pI)?
4. Which AA are chiral?
5. The nature of the side chains drastically affects the structure and function. Some are charged + or - and some are polar or nonpolar. The chart on p. 613 shows this.
6. Know the oxidation of Cysteine to form a disulfide bond (a covalent bond.) p. 616
7. What is a peptide bond? How is it formed? What general organic structure is it?
8. Be able to draw any polypeptide using p. 613. (the polypeptide waltz: $\text{N}-\text{C}_\alpha-\text{C}-\text{N}-\text{C}_\alpha-\text{C}$)
9. Know some examples of proteins and examples of the functions that proteins fulfill (p. 623).
10. How are proteins classified? What is a prosthetic group (p. 625)? Give examples.
11. Study thoroughly the levels of structural complexity of proteins (primary, secondary, tertiary, and quaternary) and what kinds of chemical forces hold them together.
12. What does it mean to say that a protein is denatured? How does that differ from a protein which is coagulated? What causes these changes?